Distribution Flash Cards

GENERAL ASSEMBLY

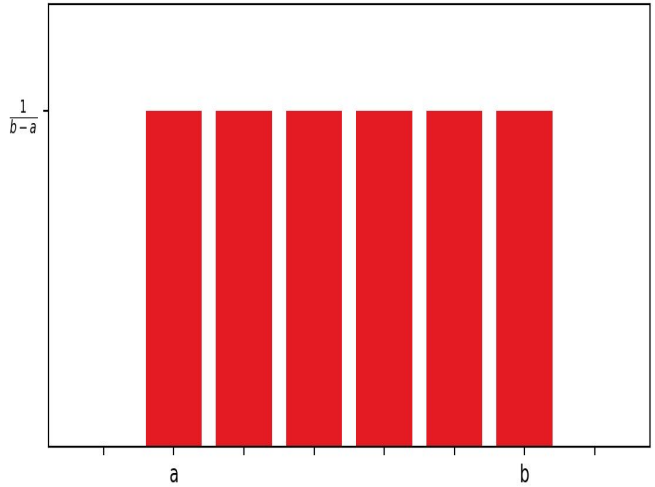
Discrete Distributions

The Discrete Uniform Distribution

Used when all outcomes are equally likely.
Ex: Outcome of rolling a six-sided die might be U(1, 6)

pmf	$f(x) = \frac{1}{b-a}$
support	$a \leq x \leq b$
mean	$\frac{a+b}{2}$
variance	$\frac{(b-a)^2}{12}$

$$X \sim U(a, b)$$



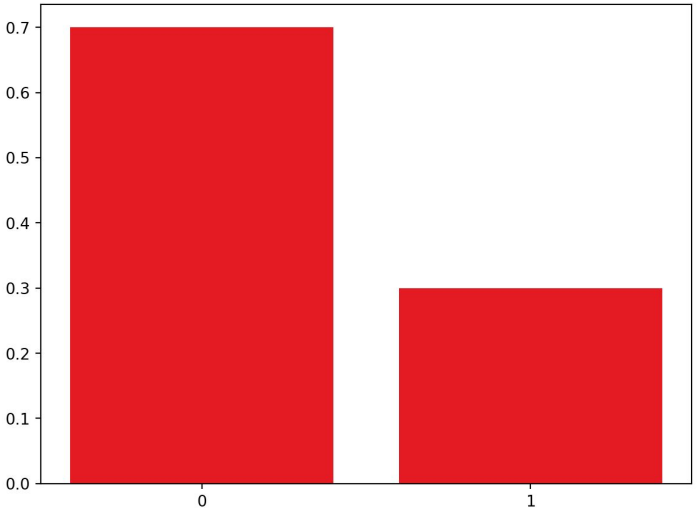
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The Bernoulli Distribution

Used when there are only two outcomes, and the “success” has fixed probability *p*
Ex: Outcome of flipping a fair coin is Ber(0.5)

pmf	$f(x) = p^x(1-p)^{1-x}$
support	$x \in \{0, 1\}$
mean	p
variance	$p(1-p)$

$$X \sim \text{Ber}(p)$$



This graph depicts Ber(0.3)

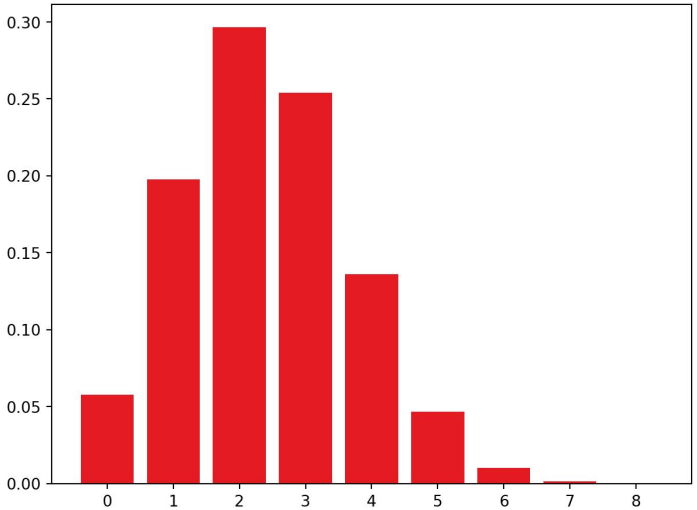
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The Binomial Distribution

Used when counting the number of successes when conducting *n* Bernoulli trials with fixed probability *p*.
Ex: Number of heads when flipping a fair coin 10 times is Bin(10, 0.5)

pmf	$f(x) = \binom{n}{k} p^x (1-p)^{n-x}$
support	$0 \leq x \leq n$
mean	np
variance	$np(1-p)$

$$X \sim \text{Bin}(n, p)$$



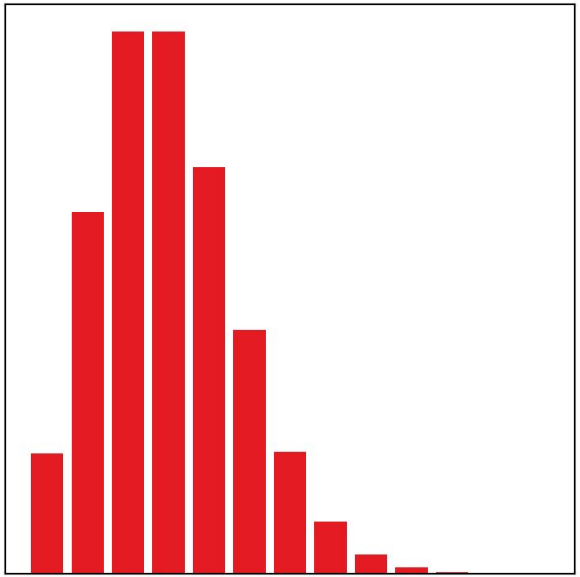
This graph depicts Bin(8, 0.3)

The Poisson Distribution

Used when estimating counts in a fixed interval (usually of time).
Ex: If you generally receive 5 emails a day, you might model this distribution with Poi(5).

pmf	$f(x) = \frac{\lambda^x e^{-\lambda}}{x!}$
support	$x \geq 0$
mean	λ
variance	λ

$$X \sim \text{Poi}(\lambda)$$



Discrete Extras

Still valuable to know, but not as frequently used or relevant

The Geometric Distribution

Suppose you have a repeated Bernoulli trial event with probability of success *p*. The **geometric distribution** describes how many trials you have until your first success.

pdf	$f(x) = (1-p)^{x-1}p$
support	$x \geq 1$
mean	$\frac{1}{p}$
variance	$\frac{1-p}{p^2}$

Example: If ⅕ candies in a pack are green, the number of candies you eat until you randomly eat a green one might be Geometric(⅕).

The Hypergeometric Distribution

Suppose you take samples from a fixed population where

- N = population size
- K = successes in population
- n = sample size

Then the number of successes you see in your sample might follow Hypergeom(N, K, n).

Example: In a game of poker, you draw 5 cards. The number of aces in your hand would follow Hypergeom(N = 52, K = 4, n = 5).

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The Multinomial Distribution

Consider a special Bernoulli trial in which there were more than two possible outcomes. Because we're not dealing with just 0s and 1s anymore, we have our first **random vector**!

Example: Suppose it's known the American population's voting registry is 40% Democrat, 40% Republican, and 20% Independent/Other. If we go out and sample 100 people, the distribution of our results might be:

$$\mathbf{X} \sim \text{Multinomial}\left(100, \begin{bmatrix} 0.4 \\ 0.4 \\ 0.2 \end{bmatrix}\right)$$

Continuous Distributions

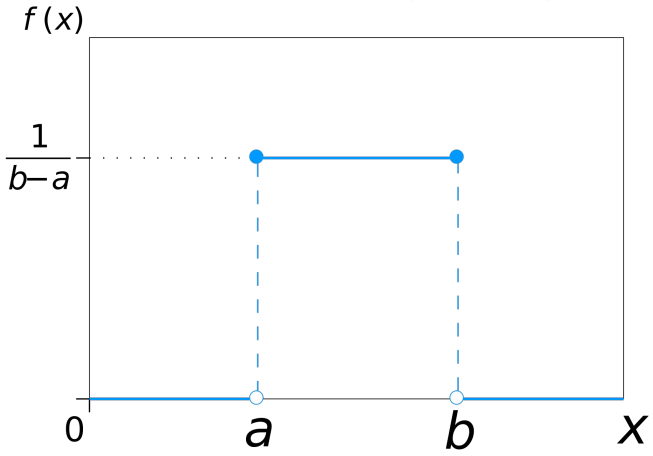


The Continuous Uniform

If all numbers in a *continuous* interval [*a*, *b*] are equally likely, you could use a **continuous uniform distribution** to model this scenario.

pdf	$f(x) = \frac{1}{b-a}$
support	$a \leq x \leq b$
mean	$\frac{a+b}{2}$
variance	$\frac{(b-a)^2}{12}$

$$X \sim U(a, b)$$

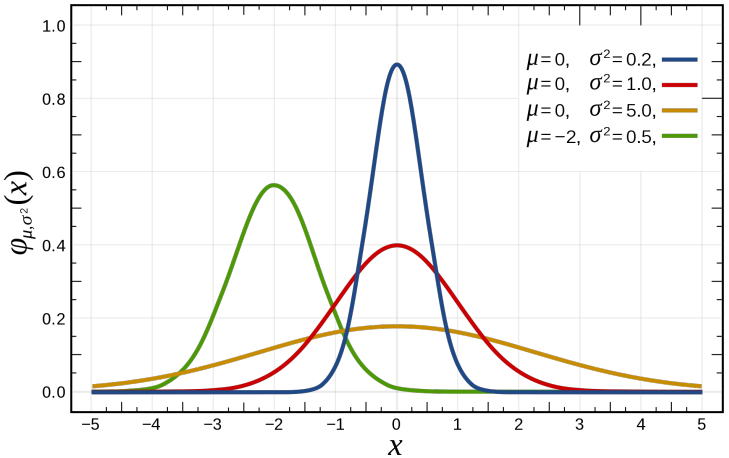


Stolen unapologetically from Wikipedia.

The Normal Distribution

pdf	$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}}$
support	$-\infty < x < \infty$
mean	μ
variance	σ^2

$$X \sim N(\mu, \sigma^2)$$



Stolen unapologetically from Wikipedia.

The Standard Normal Distribution

There's a special case of the normal distribution called the **standard normal distribution**, which occurs when $\mu = 0$, $\sigma = 1$. It can be attained by standardizing the normal distribution, just like we did when standardizing data the other day!

$$X \sim N(\mu, \sigma^2)$$

$$Z = \frac{X - \mu}{\sigma}$$

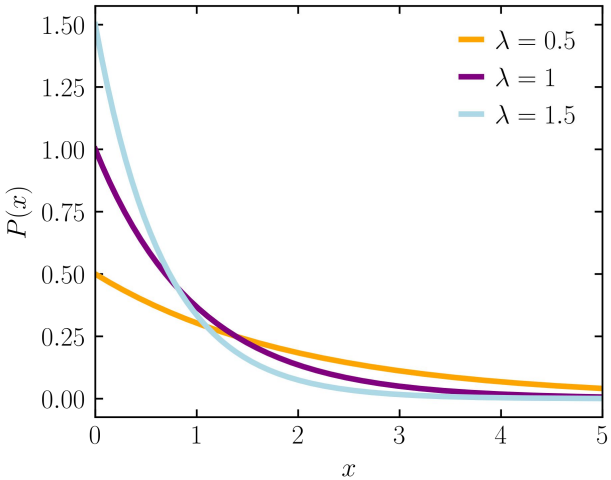
$$\implies Z \sim N(0, 1)$$

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The Exponential Distribution (mean notation)

pdf	$f(x) = \frac{1}{\beta} e^{-x/\beta}$
support	$x > 0$
mean	β
variance	β^2

$$X \sim \text{Exp}(\beta)$$

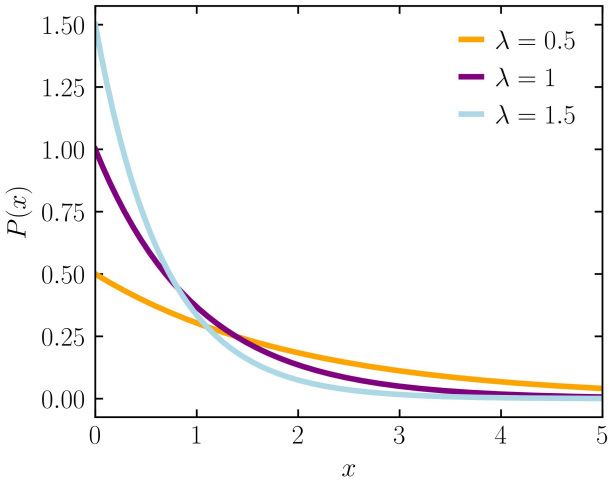


Stolen unapologetically from Wikipedia.

The Exponential Distribution (rate notation)

pdf	$f(x) = \lambda e^{-\lambda x}$
support	$x > 0$
mean	$\frac{1}{\lambda}$
variance	$\frac{1}{\lambda^2}$

$$X \sim \text{Exp}(\lambda)$$



Stolen unapologetically from Wikipedia.

Continuous Extras

Still valuable to know, but not as frequently used or relevant

The Gamma Distribution

It's very similar to the exponential distribution except with an additional **shape** parameter, α. Two interesting gamma facts that are worth knowing:

- Just like exponential, it has support $x > 0$, is skew right, and is used to model waiting times.
- ... Actually, it's used to model α consecutive waiting times! That is... if we have α exponentials:

$$X_1 + \cdots + X_\alpha \sim \text{Gamma}(\alpha, \beta)$$

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The Beta Distribution

The Beta distribution is rarely seen outside the context of Bayesian statistics. Here are some facts about it:

- Has two parameters, α and β, and in practice they're usually the same number. They control how "bowl" or "hill" shaped the pdf is.
- Since the support is between 0 and 1, the Beta distribution is often used to model probabilities in **Bayesian statistics**, which we'll get to much later.
- The standard uniform is a special case: Beta(1, 1) = U(0, 1)
- We can take a look at some cool Beta pics here.